Sunburst charts are an effective tool for visualising hierarchical data, as they provide a comprehensive view of the data structure compactly and intuitively. In addition, interactive features such as zooming and drill-down allow users to explore specific portions of the hierarchy in detail while maintaining the overall context.

After reading the articles by Woodburn et al. (2020) and Walker (2018), several advantages and disadvantages of sunburst charts can be identified.

The advantages of using sunburst charts are:

- Efficient use of the space for deep hierarchies, as the radial representation leaves more room for the inner nodes.
- Clear hierarchical representation since the structure enables users to perceive the hierarchical level in an intuitive way.
- Effective encoding of quantitative data because the angular width represents a quantitative variable. In addition, the colour encoding and the structural organisation help identify extreme values, clusters, and good/bad performance areas.
- Zoom-in-place functionality ensures that the current level remains visible during navigation, preserving context.
- Implicit Breadcrumb navigation, because the selected node remains visible during zooming
- The nodes are not removed, ensuring consistent animations.

The disadvantages are:

- Label readability issues because the text orientation can be less consistent compared to other types of visualisation.
- Navigation issues in a non-interactive setting, where exploring large hierarchies becomes difficult without zooming capabilities.
- The chart does not explicitly show the dominance relationships.
- Limited effectiveness in dynamic contexts, since adding new elements can alter the overall structure.

If interactive facilities were not available, a good solution would be the treemaps, as discussed in Walker's article. Treemaps represent hierarchical data using nested rectangles and encode quantitative values through area, while maintaining good readability.

Reference list

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